GATE ME 90 Days Challenge

Day 57/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
General Aptitude	Spatial Aptitude	45 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on mirror and water images of figures, words and clocks, rotation of figures, paper folding and cutting (one fold, two folds, diagonal folds), dice and cube nets (opposite faces, possible cubes), figure series, counting triangles and squares, painted-cube problems and shortest paths on solids. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A plane mirror is held vertically to the right of the word GATE90, as shown. Which option is its mirror image?



(A) 

(B) 

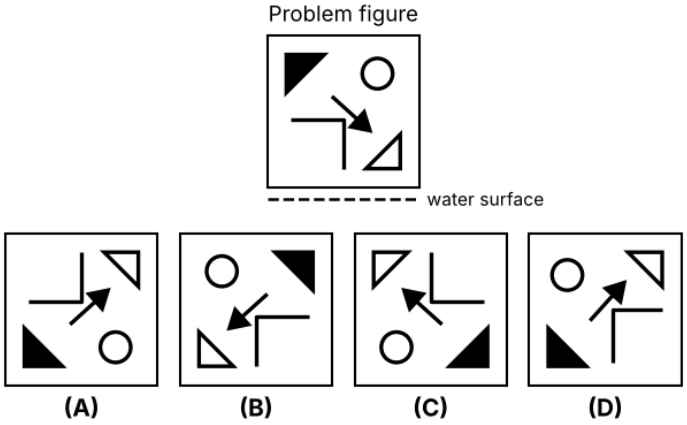
(C) 

(D) 

(A) Figure (A)
(B) Figure (B)
(C) Figure (C)
(D) Figure (D)

Q2 MCQ

Which option is the water image of the problem figure (water surface below the figure)?



(A) Figure (A)
(B) Figure (B)
(C) Figure (C)
(D) Figure (D)

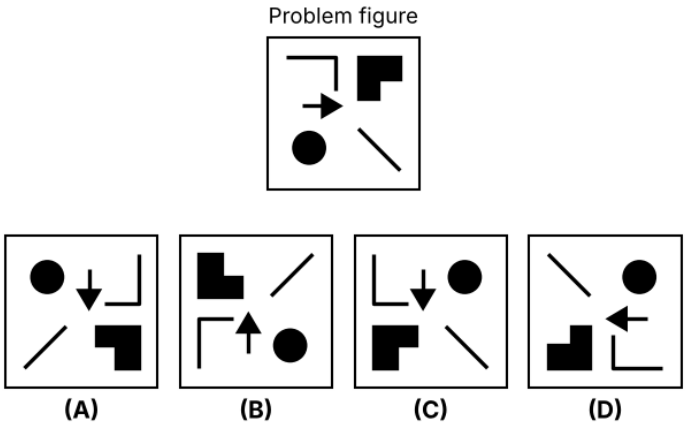
Q3 MCQ

The actual time on a wall clock is 8:15. What time does its image in a plane mirror (held vertically beside it) appear to show?

(A) 4:45
(B) 3:15
(C) 9:45
(D) 3:45

Q4 MCQ

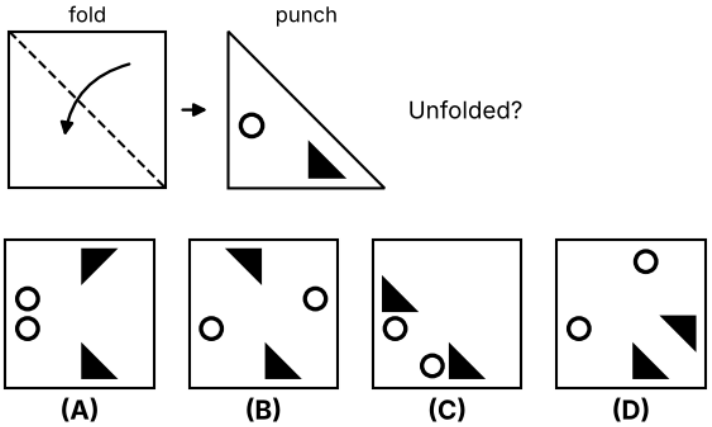
Which option shows the problem figure rotated through 90° anticlockwise about its centre?



(A) Figure (A)
(B) Figure (B)
(C) Figure (C)
(D) Figure (D)

Q5 MCQ

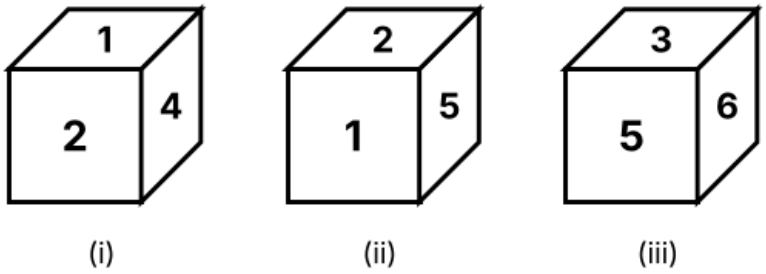
A square sheet is folded along its diagonal (the upper-right half over the lower-left half) and a circle and a triangle are punched through it as shown. Which option shows the unfolded sheet?



(A) Figure (A)
(B) Figure (B)
(C) Figure (C)
(D) Figure (D)

Q6 MCQ

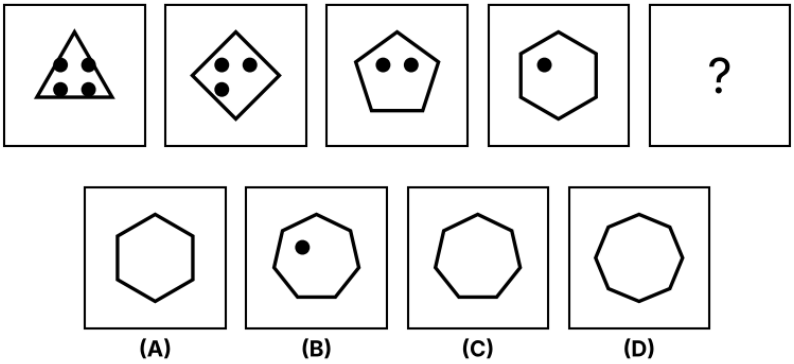
Three views of the same die (faces numbered 1 to 6) are shown. Which number is on the face opposite to 4?



(A) 3
(B) 5
(C) 2
(D) 6

Q7 MCQ

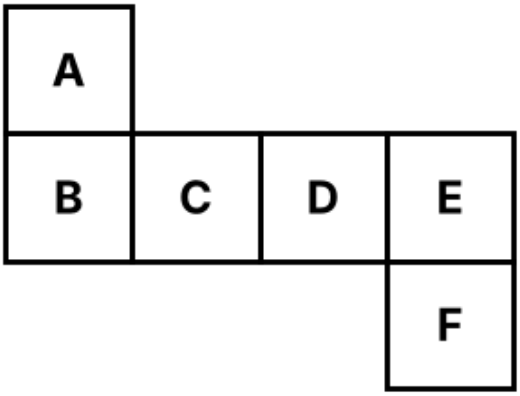
Which option figure continues the series?



(A) Figure (A)
(B) Figure (B)
(C) Figure (C)
(D) Figure (D)

Q8 MCQ

The net shown is folded into a cube. Which pair of faces will be opposite each other?



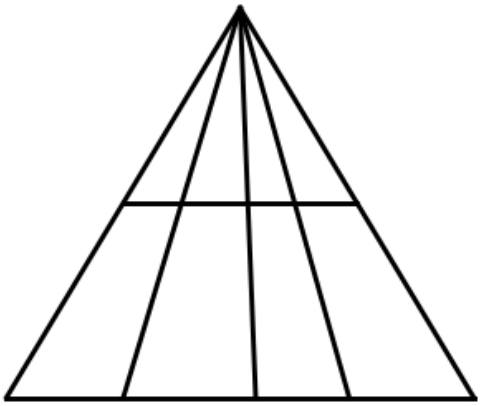
(A) A and C
(B) B and E
(C) C and E
(D) D and F

Q9 NAT

A wooden cuboid 5 cm × 4 cm × 3 cm is painted on all its faces and cut into 1 cm cubes. The number of cubes with exactly two painted faces is _____

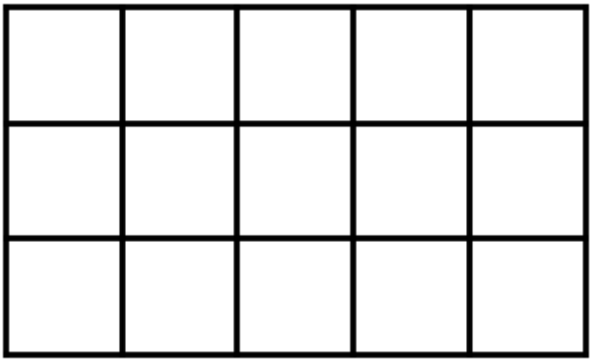
Q10 NAT

The number of triangles in the figure (a triangle with three lines from the apex to the base and one line parallel to the base) is _____



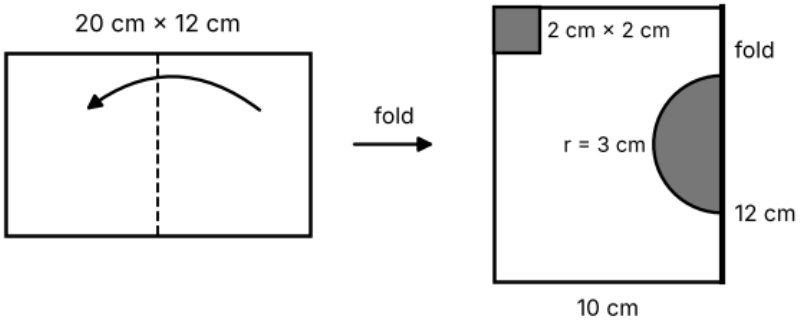
Q11 NAT

The number of squares of all sizes in the grid of 3 × 5 unit squares shown is _____



Q12 NAT

A rectangular sheet 20 cm × 12 cm is folded in half to give a 10 cm × 12 cm piece. A semicircle of radius 3 cm centred on the middle of the fold, and a 2 cm × 2 cm square at a free corner, are cut out as shown. When the sheet is unfolded, the area of the remaining paper (in cm², round off to two decimal places) is _____



Q13 NAT

An ant moves on the outer surface of a cube of side 10 cm from a corner P to the diagonally opposite corner Q. The length of the shortest path (in cm, round off to two decimal places) is _____

Q14 NAT

A rectangular sheet 44 cm × 21 cm is rolled along its longer side so that the 44 cm edge forms the circumference of an open cylinder 21 cm tall. The volume of the cylinder (in cm³, round off to one decimal place) is _____

Q15 NAT

A 3 × 3 × 3 cube is built from 27 unit cubes. A square tunnel one cube wide is cut straight through the centre of each face to the opposite face (three tunnels in all). The number of unit cubes that remain is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(C) Figure (C)
Q2	MCQ	(A) Figure (A)
Q3	MCQ	(D) 3:45
Q4	MCQ	(B) Figure (B)
Q5	MCQ	(D) Figure (D)
Q6	MCQ	(B) 5
Q7	MCQ	(C) Figure (C)
Q8	MCQ	(C) C and E
Q9	NAT	24
Q10	NAT	20
Q11	NAT	26
Q12	NAT	203.73 (accept 202.70 to 204.75)
Q13	NAT	22.36 (accept 22.20 to 22.50)
Q14	NAT	3235.3 (accept 3219.1 to 3251.5)
Q15	NAT	20

Detailed Solutions

Q1 **Answer: (C) Figure (C)**

Concept: A vertical mirror reverses left and right only; top and bottom stay where they are.

So the order of the characters is reversed (0 9 E T A G) *and* every character is turned left-to-right: **(C)**.

(A) is the water image (upside down, same order), (B) reverses each letter but keeps the order, (D) is a 180° rotation.

Exam tip: Letters with a vertical axis of symmetry (A, T, 0) look unchanged in a mirror; check the asymmetric ones (G, E, 9) to decide.

Q2 **Answer: (A) Figure (A)**

Concept: A water image reflects in a horizontal line: top and bottom interchange, left and right stay.

The filled triangle at the top-left goes to the bottom-left, the circle at the top-right to the bottom-right, the open triangle at the bottom-right to the top-right, and the arrow that pointed down-right now points up-right: **(A)**.

(B) is the mirror image (left-right), (C) a 180° rotation and (D) a 90° rotation.

Exam tip: Track one asymmetric element (here the arrow) first; it usually eliminates two options at once.

Q3 **Answer: (D) 3:45**

Concept: A mirror reflects each hand about the 12–6 line, so the apparent time = 11:60 − actual time (for times between 1:00 and 11:00).

11:60 − 8:15 = **3:45**: **(D)**.

Check: at 8:15 the minute hand is at 3; reflected about the 12–6 line it points to 9, i.e. 45 minutes. The hour hand just past 8 appears just before 4, i.e. between 3 and 4.

Exam tip: For the water image of a clock, the 12–6 line becomes a 3–9 line; use 18:30 − time instead.

Q4 **Answer: (B) Figure (B)**

Concept: In a 90° anticlockwise turn: top-right → top-left → bottom-left → bottom-right → top-right; directions turn the same way (right → up).

The filled L at the top-right moves to the top-left, the open corner at the top-left moves to the bottom-left, the dot to the bottom-right, the slash to the top-right, and the arrow now points up: **(B)**.

(A) is a 90° clockwise turn, (C) a mirrored figure turned 90°, (D) a 180° turn.

Exam tip: In a rotation the figure is never reversed: if the shape of any element is flipped, the option is a reflection, not a rotation.

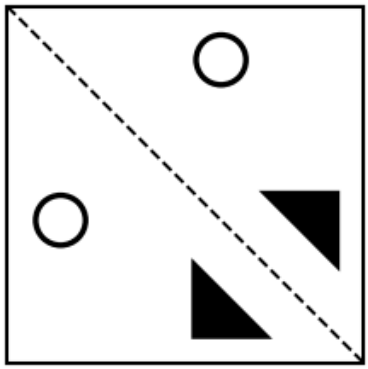
Q5 **Answer: (D) Figure (D)**

Concept: On unfolding, each hole is reflected in the fold line; here the fold line is the diagonal from the top-left to the bottom-right corner.

Reflection in this diagonal maps a point (x, y) to (−y, −x): the circle near the left edge reappears near the top edge, and the triangle near the bottom edge reappears near the right edge, turned so that it is the mirror image about the diagonal: **(D)**.

(A) reflects in the horizontal centre line, (B) rotates through 180°, (C) reflects in the other diagonal.

Exam tip: The unfolded pattern is always symmetric about every fold line; check that first.



Q6 **Answer: (B) 5**

Concept: Faces seen together in one view are adjacent, so they cannot be opposite; each face has exactly four neighbours.

Face 5 appears with 2 and 1 in view (ii) and with 3 and 6 in view (iii); so 5 is adjacent to 1, 2, 3 and 6.

The only face left is 4, so 4 is opposite 5: **(B)**.

Exam tip: Start with the number that occurs in the most views; listing its neighbours often gives the answer without any rotation.

Q7 **Answer: (C) Figure (C)**

Concept: Look for two independent changes from one figure to the next.

The polygon gains one side at each step (3, 4, 5, 6 → 7 sides) and the number of dots falls by one (4, 3, 2, 1 → 0).

Next figure: a heptagon with no dot: **(C)**.

Exam tip: Count sides carefully; a hexagon (A) and an octagon (D) are the usual traps.

Q8 **Answer: (C) C and E**

Concept: In a straight row of faces in a net, alternate faces (one apart) are opposite. Faces sharing an edge are never opposite.

Row B–C–D–E: B is opposite D, and C is opposite E: **(C)**.

The remaining pair, A and F, are opposite: they are flaps on opposite sides of the row, so one becomes the top and the other the bottom.

Exam tip: B and E, the ends of a row of four, meet when the row wraps round, so they are adjacent, not opposite.

Q9 **Answer: 24**

Concept: Cubes with exactly two painted faces lie along the edges, excluding the corner cubes.

Each dimension has 4 edges; the cubes on one edge, without its two corners, number (length − 2).

$4[(5-2)+(4-2)+(3-2)] = 4 \times 6 =$ **24**

Exam tip: Also: 8 corner cubes (three faces), $2[(3\times2)+(3\times1)+(2\times1)] = 22$ with one face, and $3 \times 2 \times 1 = 6$ unpainted; total $60 = 5 \times 4 \times 3$.

Q10 **Answer: 20**

Concept: With n lines from the apex (including the two sides), any two of them together with a transversal form a triangle.

Here n = 5 lines from the apex, so each transversal gives $C(5, 2) = 10$ triangles.

There are two transversals (the base and the parallel line): $2 \times 10 =$ **20**

Exam tip: For k transversals the count is $k \times C(n, 2)$, provided no other lines cross inside.

Q11 **Answer: 26**

Concept: In an $m \times n$ grid ($m \le n$), the number of $k \times k$ squares is $(m-k+1)(n-k+1)$.

1×1: $3 \times 5 = 15$; 2×2: $2 \times 4 = 8$; 3×3: $1 \times 3 = 3$

Total $= 15 + 8 + 3 =$ **26**

Exam tip: Rectangles (all sizes) would be $C(4, 2) \times C(6, 2) = 6 \times 15 = 90$.

Q12 **Answer: 203.73 (accept 202.70 to 204.75)**

Concept: Each cut is reflected in the fold line on unfolding; a cut touching the fold becomes one symmetric hole.

Semicircle on the fold → a full circle of radius 3 cm: area $9\pi = 28.27$ cm²

Corner square → two squares at the two corners of that long edge: $2 \times 4 = 8$ cm²

Remaining $= 240 - 28.27 - 8 =$ **203.73 cm²**

Exam tip: The free corner of the folded piece is two layers thick, so it gives two holes, not four.

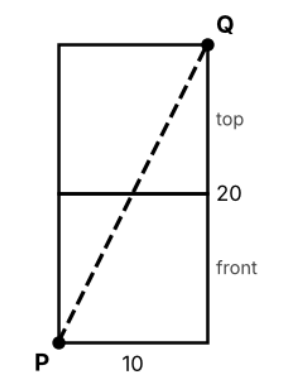
Q13 **Answer: 22.36 (accept 22.20 to 22.50)**

Concept: Unfold two adjacent faces into one plane; the shortest surface path becomes a straight line.

The unfolded rectangle is 10 cm × 20 cm, and P and Q are its opposite corners (see figure).

Path = $\sqrt{(10^2 + 20^2)} = \sqrt{500} =$ **22.36 cm** (the space diagonal 17.32 cm is not on the surface).

Exam tip: The path crosses the common edge at its midpoint; there are six such shortest paths.



Q14 **Answer: 3235.3 (accept 3219.1 to 3251.5)**

Concept: When a sheet is rolled, the rolled edge becomes the circumference and the other edge the height.

$2\pi r = 44 \rightarrow r = 7.003$ cm ($r = 7$ with $\pi = 22/7$)

$V = \pi r^2 h = 44^2 \times 21/(4\pi) =$ **3235.3 cm³** (3234 with $\pi = 22/7$, inside the accepted range)

Exam tip: Rolling along the shorter side gives $r = 21/(2\pi)$ and $h = 44$, i.e. $V = 21^2 \times 44/(4\pi) = 1544$ cm³, a different answer.

Q15 **Answer: 20**

Concept: Each tunnel removes the centre column of 3 cubes, but the very centre cube is common to all three tunnels.

Removed = the 6 face-centre cubes + the 1 body-centre cube = 7 (that is, 3 + 2 + 2 by inclusion–exclusion).

Remaining = 27 − 7 = **20** (the 8 corners and 12 edge cubes)

Exam tip: This is the first stage of the Menger sponge.

What next?

Day 57 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 58/90: Manufacturing, Joining, Powder Metallurgy & Polymers. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.